

3] filter function - filter function in a python is a useful function, that applies a given function to each item of an iterable (list, set, tuple) & we filter the items on the basis of if condition.

map -

numbers = [1, 2, 3, 4, 5] ← list (iterable) ⑤

```
def square(x):  
    return x**2
```

← function

result = map(square, numbers)

result1 = list(result)

print(result)

→ [1, 4, 9, 16, 25] ⑤

$$1 \times 1 = 1^2 = 1$$

$$3 \times 3 = 3^2 = 9$$

$$2 \times 2 = 2^2 = 4$$

$$4 \times 4 = 4^2 = 16$$

$$5 \times 5 = 5^2 = 25$$

Syntax :-

`filter (function, iterable)`

filter function returns true or false, for elements where condition is true, those elements are available in the final output

eg. filter out odd elements,
 & keep even elements using
 filter.

numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

```
def is_even(x):
    return x % 2 == 0
```

← fun

```
return x % 2 == 0
```

$$2\%, 2 \approx 0$$

```
result = filter('is-even', numbers)
```

```
count1 = list(result)
```

result = list(result)

print(result) → [2, 4, 6, 8, 10]

using lambda function:- 10 elements

numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

F T F T F T F T F T
 $2 \% 2 == 0$

even_numbers = filter(lambda x: x % 2 == 0, numbers)

result = list(even_numbers)

print(result) → [2, 4, 6, 8, 10]

5 element

using filter function, create a new list where elements are less than equal to 5.

numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

def fun1(x):

K=5 → True

K=5 → False

return $x \leq 5$

result = list(filter(fun1, number))

print(result) $\rightarrow [1, 2, 3, 4, 5]$

4] reduce function -

1) reduce function takes first two items of iterable (list, tuple, set, array, str)

& applied to function, next we take o/p of function & next item in the iterable & applied to function, & we do this till the end of iterable & at the end we get single cumulative value

map- iterable = [1, 2, 3, 4, 5]

def fun1()

reduce iterable = [1, 2, 3, 4, 5]

def func():
 return inf



Syntax :-

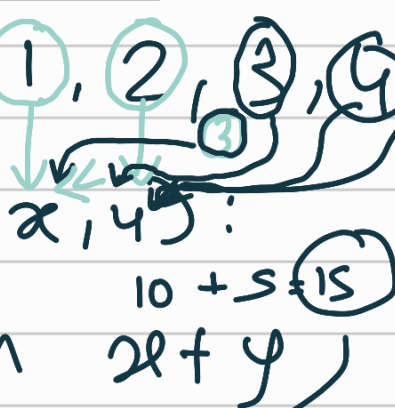
reduce (function, iterable)

eg.

import functools import reduce

numbers = [1, 2, 3, 4, 5]

def add (x, y):
 10 + 5 = 15
 return x + y



result = reduce (add, numbers)

print (result) → 15